满分：150分

116分

-10

此处未上传图片

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二填空题

13. $\frac{1}{3n-2}$

14. $[-1,2]$

15. 2

16. $\frac{2021}{2}$

17.

解：(1) $2a-a\cos B=b\cos A$

据正弦定理

$2\sin A-(\sin A\cos B+\sin B\cos A)=0$

$2\sin A-\sin C=0$

$2\sin A=\sin C$

$\frac{\sin A}{\sin C}=\frac{1}{2}$

据正弦定理，$2a=c$

$\frac{a}{c}=\frac{1}{2}$

(2) $\cos C=\frac{a^2+b^2-c^2}{2ab}=\frac{1}{4}$

将 $2a=c$ 及 $b=4$ 代入

解得 $a=2$，$c=4$

$\sin C=\sqrt{1-\cos^2 C}=\frac{\sqrt{15}}{4}$

$\therefore S_{\triangle ABC}=\frac{1}{2}ab\sin C=\sqrt{15}$

18.

解：(1) $\begin{cases} a_1+5d-a_1-d=8 \\ (a_1+5d)^2=a_1(a_1+24+20d) \end{cases}$

$\therefore a_1=5$，$d=2$

$\therefore \{a_n\}$ 为首项为5，公差为2的等差数列

$\therefore a_n=3+2n$，$n\in N^*$

(2)

$$b_n = \frac{1}{(3+2n)(2n+5)} = \cancel{2n+3=2n+}$$

$$= \frac{1}{2}\left(\frac{1}{2n+3} - \frac{1}{2n+5}\right)$$

$$S_n = \frac{1}{2}\left(\frac{1}{5} - \frac{1}{7}\right) + \frac{1}{2}\left(\frac{1}{7} - \frac{1}{9}\right) + \cdots + \frac{1}{2}\left(\frac{1}{2n+3} - \frac{1}{2n+5}\right)$$

$$= \frac{1}{2}\left(\frac{1}{5} - \frac{1}{2n+5}\right)$$

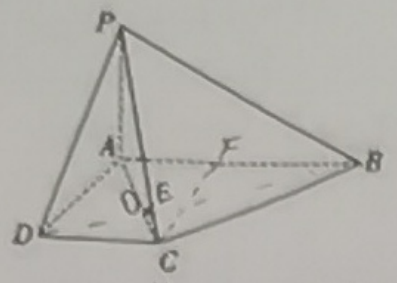
$$= \frac{1}{10} - \frac{1}{4n+10}$$

令 $S_n = \frac{1}{11}$

则 $\frac{1}{10} - \frac{1}{4n+10} = \frac{1}{11}$

$\therefore n = 25$

19.



解：(1) 证明：作 $CF \parallel AD$，交 AB 于 F

又 $\because AB \parallel CD$

$\therefore$ 四边形 $ADCF$ 为平行四边形

$\therefore AF = DC = 1$，$FB = AB - AF = 1$

$\therefore F$ 为 AB 中点，$CF = AD = 1$

$\therefore$ 直角梯形 $ABCD$

$\because AD \perp AB$，$AD \perp DC$ $\therefore CF \perp AB$

$\therefore$ 根据勾股定理，$AC = \sqrt{2}$

$BC = \sqrt{2}$，

在 $\triangle ABC$ 中，$AB^2 = 4$，$AC^2 + BC^2 = 4$

$\therefore AC^2 + BC^2 = AB^2$，$\triangle ABC$ 为直角三角形

$\therefore BC \perp AC$

$\because PA \perp$ 平面 $ABCD$，$BC \subset$ 平面 $ABCD$

$\therefore PA \perp BC$

又 $\because BC \perp AC$，AC，PAC 平面 PAC

$PA \cap AC = A$

$\therefore BC \perp$ 平面 PAC 又 $BC \subset$ 平面 PBC

$\therefore$ 平面 $PBC \perp$ 平面 PAC

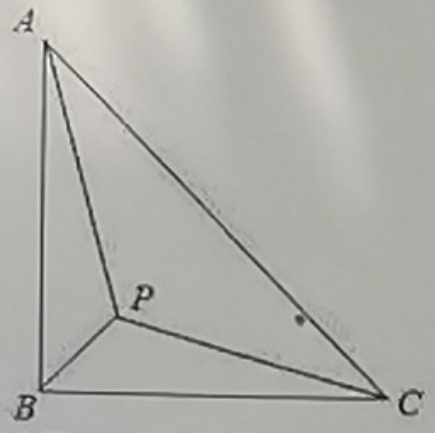
(2) 连接 BD，与 AC 交于 O

从 O 向 PC 做垂线 $OE \perp PC$

则 OE 为 O 点到 PBC 的距离

则 D 到 PBC 距离为

20.



解:(1) ~~∵∠ABP=∠APB=~~

∵ $\angle ABP=45^\circ$, $\angle APB=120^\circ$

∴ $\angle BAP=180^\circ-\angle ABP-\angle APB=15^\circ$

又∵ $\angle BAC=45^\circ$

∴ $\angle PAC=\angle BAC-\angle BAP=30^\circ$

∵ $\angle BPC=120^\circ$

∴ $\angle PBC+\angle PCB=60^\circ$

又∵ $\angle PBA+\angle BAC+\angle ACP+\angle PCB+\angle PBC=180^\circ$

∴ $\angle PBC+\angle PCB+\angle ACP=90^\circ$

∴ $\angle ACP=30^\circ$

又∵ $\angle PAC=30^\circ$

∴ $AP=CP=2$

∴ $S_{\triangle APC}=\frac{1}{2}AP\cdot CP\cdot\sin\angle APC=\sqrt{3}$

(2) $\sin 15^\circ=\sin(45^\circ-30^\circ)$

$=\sin 45^\circ\cos 30^\circ-\sin 30^\circ\cos 45^\circ$

$=\frac{\sqrt{2}}{2}\times\frac{\sqrt{3}}{2}-\frac{1}{2}\times\frac{\sqrt{2}}{2}$

$=\frac{\sqrt{6}-\sqrt{2}}{4}$

根据正弦定理

$$\frac{PA}{\sin\angle ABP}=\frac{PB}{\sin\angle BAP}$$

$\therefore PB = \sin 15^\circ \cdot \frac{PA}{\sin 45^\circ}$

$= \sqrt{3} - 1$

-4

21.

解：(1) $d=2$

$(a_1+3d)^2 = (a_1+d)(a_1+7d)$

$\therefore a_1 = 2$

$\therefore a_n = \cancel{2^{n-1}}\ 2n,\ n \in N^*$

$\therefore$ ~~$b_2 = a_2 = 2^2$~~

~~$b_3 = a_4 = 2^4$~~

~~$b_4 = a_8 = 2^8$~~

~~$\therefore b_n = 2^{2^{n-1}} = 4^{n-1}$~~

$\therefore q = \frac{b_3}{b_2} = 2$

$\therefore b_n = \cancel{b_2(n-2)4}\ b_2 \cdot q^{n-2} = 2^n$

~~$b_2 4^{n-2} = 4^{n-1}$~~

$\therefore$ 综上，$a_n = 2n,\ n \in N^*$

$b_n = 2^n,\ n \in N^*$

(2) $\frac{c_1}{2} + \frac{c_2}{4} + \frac{c_3}{6} + \frac{c_4}{8} + \cdots + \frac{c_n}{2n} = 2^{n+1}$

假设 $c_n = \begin{cases} \cancel{2}\ 8, & n=1 \\ 2n \cdot 2^n, & n \geq 2 \end{cases}$

则原式 $= 2 + 2 + 4 + 8 + \cdots + 2^n$

$= 2 + \frac{2(1-2^n)}{1-2} = 2 - 2 + 2^{n+1} = 2^{n+1} =$ 右

$\therefore$ 假设成立

~~$\therefore S_n$ 令 S_n 为 $\{c_n\}$ 的前 n 项和~~

~~$\therefore S_n =$~~

$S_{2020} = 4 + 2\times2^1 + 2\times2\times2^2 + 2\times3\times2^3 + \cdots + \cancel{2n\cdot2}\ 2\times2020\times2^{2020}$

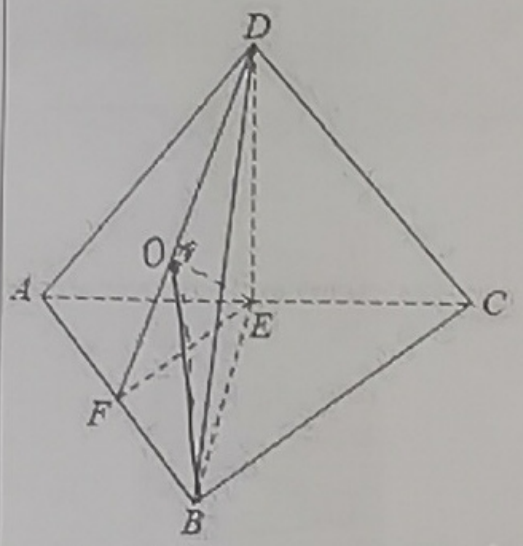
$\therefore 2S_{2020} = 8 + 2\times2^2 + 2\times2\times2^3 + 2\times3\times2^4 + \cdots + 2\times2020\times2^{2021}$

$\therefore S_{2020} = 2S_{2020} - S_{2020} = 2\times2020\times2^{2021} - 2(2^2+2^3+2\cdots+2^{2020})$

$= 4038\times2^{2021} + 8$

22.

-8



连接OB，

则∠EBO即为EB与平面ABD所成角

解：(1) ∵ E为D在ABC上的射影

∴ DE⊥平面ABC，又AC⊂ABC

∴ DE⊥AC

∵ AD=CD

∴ E为AC中点

∵ DF⊥AB，DA=DB

∴ F为AB中点

∴ EF∥BC，EF=½BC

又∵ EF⊂平面EFD

BC⊄平面EFD

∴ BC∥平面DEF

(2) ∵ ∠ADC = $\frac{\pi}{3}$

又∵ AD=DC，DE⊥AC

∴ ∠ADE = ∠CDE = $\frac{\pi}{6}$

∴ ∠DAE = $\frac{\pi}{6}$ = ∠PCE

∴ AD=DC=AC

作EO⊥DF于O

则 ~~EO为~~ O点为E在平面ABD
上的射影